Here, $\hat{\tau}_h$ for $h < 0$ represents "placebo" pre-treatment effect estimates as in Equation 5 and $\hat{\tau}_h$ for $h \geq 0$ represents the intended post-treatment ATT_h estimates. $h = -1$ is intentionally omitted from the regression to serve as the counterfactual baseline.

While the early econometric literature extensively uses the two-way fixed effect (TWFE) estimator for DiD studies, it is important to note that the OLS-estimated $\hat{\beta}$s (Equation 8) and $\hat{\tau}_h$s (Equation 9) do not really estimate the ATT and ATT_h defined in Section 3.1.3. Instead, under both the parallel trend assumption and the treatment effect homogeneity assumption (i.e., the treatment effect does not vary over time and adoption cohorts), a TWFE estimator equals a variance weighted average of all possible two-group/two-period DiD estimators in the data [54]. However, the treatment effect homogeneity assumption is a strong assumption that is likely violated in the staggered adoption setting: For example, it is reasonable to anticipate that later Cursor adoption cohorts may have stronger adoption effects as the tooling and model capabilities are rapidly advancing. If this assumption is violated, it may lead to "forbidden comparisons" and negative weighting in the TWFE estimator, biasing the estimated parameters and jeopardizing the validity of causal claims based on TWFE estimators [23, 44].

B.2 The Callaway and Sant'Anna [31] Estimator

The Callaway and Sant'Anna [31] estimator takes a fundamentally different approach from both TWFE and the Borusyak et al. [29] imputation estimator. Rather than estimating regression models on all adoption cohorts, it first estimates *group-time average treatment effects* $ATT(g,t)$—the average treatment effect for cohort g (repositories adopting in the same period) at time t—before aggregating $ATT(g,t)$ into summary ATT and ATT_h measures.

For a group of repositories g that adopt Cursor at time period g, the average treatment effect at calendar time t is identified as:

$$ATT(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G_g = 1] - \mathbb{E}[Y_t - Y_{g-1} \mid C] \tag{10}$$

where $Y_t - Y_{g-1}$ represents the evolution of the outcome from the period prior to treatment $(g-1)$ to the current period t. The researcher may choose "never-treated" repositories or "not-yet-treated" observations as the control group C; we choose the latter to align with the other two estimators. This formulation explicitly avoids using already-treated units as controls, eliminating the "forbidden comparisons" that can bias TWFE estimates [54].

With $ATT(g,t)$ estimations,

$$ATT = \sum_{g} \sum_{t \geq g} w_{g,t} \cdot ATT(g,t) \tag{11}$$

where $w_{g,t}$ are weights proportional to the size of each group for all post-treatment observations (i.e., $t \geq g$). ATT_h is given by:

$$ATT_h = \sum_{g} \sum_{t} w^h_{g,t} \cdot ATT(g,t) \tag{12}$$

where $w^h_{g,t}$ are weights proportional to the size of each group satisfying $t - g = h$. As with the other two estimators, this aggregation allows us to test the parallel trends assumption (where $h < 0$) and estimate "horizon-average" treatment effects (where $h \geq 0$).

The Callaway and Sant'Anna [31] framework provides multiple approaches to estimate $ATT(g,t)$. In our study, we use the *outcome regression* estimator with covariate adjustment. Specifically, for each group-time pair (g,t), we first estimate an outcome regression model on the comparison group C:

$$Y_{it} - Y_{i,g-1} = \alpha_{g,t} + \Gamma'_{g,t} Z_{i,g-1} + \epsilon_{it}, \quad \text{for } i \in C \tag{13}$$

where $Z_{i,g-1}$ represents pre-treatment covariates (defined in Section 3.2.2). This model estimates the expected evolution of outcomes for untreated repositories with similar pre-treatment characteristics. The estimated $ATT(g,t)$ is then computed as:

$$\widehat{ATT}(g,t) = \frac{1}{|G_g|} \sum_{i \in G_g} \left[(Y_{it} - Y_{i,g-1}) - (\hat{\alpha}_{g,t} + \hat{\Gamma}'_{g,t} Z_{i,g-1}) \right] \tag{14}$$

where G_g denotes the set of repositories in cohort g, and $(\hat{\alpha}_{g,t} + \hat{\Gamma}'_{g,t} Z_{i,g-1})$ represents the predicted counterfactual outcome change for treated repository i based on its pre-treatment characteristics. We also allow for one period of anticipation in the estimation, consistent with our treatment of the Borusyak et al. [29] estimator, where we exclude $h = -1$ from pre-trend tests.

A key distinction of the Callaway and Sant'Anna [31] estimator is that it estimates treatment effects *separately within each cohort* before aggregating. While this approach ensures clean identification by avoiding cross-cohort contamination, it can reduce statistical power when individual cohorts are small. In our setting, where many adoption cohorts contain fewer than 100 repositories (Figure 2), this cohort-specific estimation may yield noisier estimates compared to Borusyak et al. [29], which leverages all untreated observations to fit a single counterfactual outcome model.

B.3 Comparing Estimation Results

We summarize the ATT and ATT_h estimations from all three estimators in Table 6 and Figure 6. For pre-trend tests, we use the same heteroskedasticity- and cluster-robust Wald tests [29] to test the joint null hypothesis that all "placebo" pre-treatment effect estimates are equal to zero. Most models pass the pre-trend test at the 0.05 significance level. The one exception is code complexity estimated with TWFE, which shows marginally significant pre-trends visible in Figure 6. This violation likely stems from the "forbidden comparisons" inherent to the TWFE estimator [44], as both the Borusyak et al. [29] and Callaway and Sant'Anna [31] estimators show no significant pre-trends for this outcome.

For development velocity outcomes, the three estimators show qualitatively consistent results on the ATT_h estimates (Figure 6). The differences in ATT estimates stem from the fact that they are averaged differently in the three estimators. All estimators find positive effects on lines added, with estimates ranging from +28.58% (Borusyak et al. [29]) to +82.74% (TWFE) to +53.60% (Callaway and Sant'Anna [31]). The differences in manitudes stem from the fact that they use different weighted averages for the ATT estimates. While the magnitudes differ, the direction and statistical significance align, providing robust evidence that Cursor adoption increases code output. For commits, Borusyak et al. [29] and Callaway and Sant'Anna [31] find small, statistically insignificant effects (+2.63% and −0.73%, respectively), while TWFE estimates a larger, significant effect (+17.83%). This TWFE inflation likely reflects the bias from "forbidden comparisons" discussed in Section B.1.

For code quality outcomes, the estimators diverge more substantially. The Borusyak et al. [29] and TWFE estimators consistently find significant increases in static analysis warnings (+30.26% and